

MySQL

- MySQL은 오픈소스 관계형 데이터베이스 관리 시스템입니다.
- 사용자는 SQL이라는 구조화된 쿼리 언어를 사용하여 데이터를 정의, 조작, 제어, 쿼리할 수 있습니다.
- MySQL은 오픈소스이므로 25년 이상 사용자와 긴밀히 협력하여 개발한 여러 기능이 포함되어 있습니다.

설치방법

1. Docker를 이용한 설치

사전준비

- dbeaver 설치
- Docker 설치
- mysql 설치 폴더 생성
 - mkdir ./mysql
 - mkdir ./mysql/database

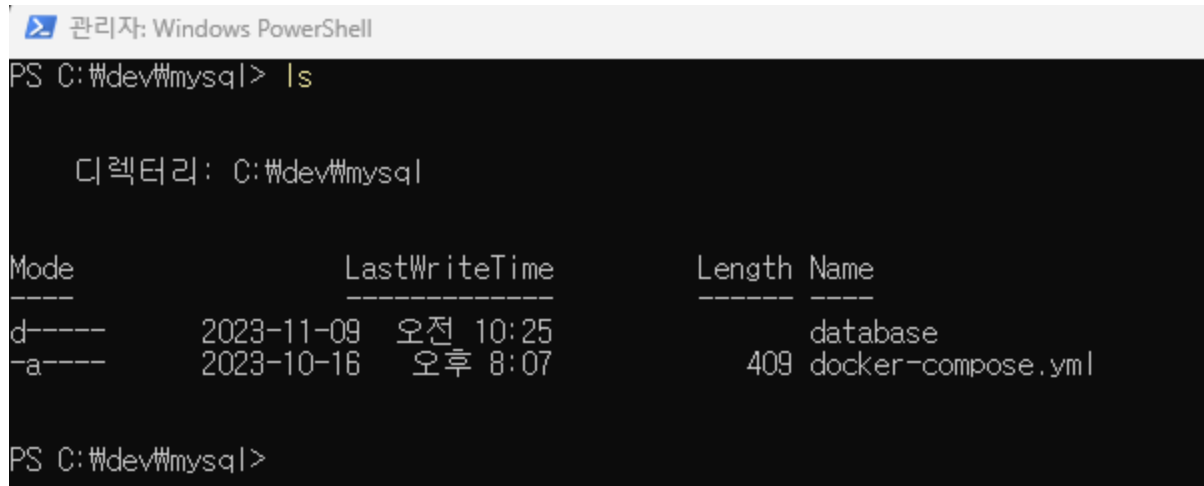
./mysql/docker-compose.yml 파일 생성

```
version: "3"

services:
  db:
    image: mysql
    restart: always
    command:
      - --character-set-server=utf8mb4
      - --collation-server=utf8mb4_unicode_ci
    volumes:
      - ./database:/var/lib/mysql
    environment:
      MYSQL_ROOT_PASSWORD: "root1234"
      MYSQL_DATABASE: "examplesdb"
      MYSQL_USER: "urstory"
      MYSQL_PASSWORD: "u1234"
    ports:
      - "3306:3306"
```

Docker를 이용한 MySQL 설치

> cd ./mysql # docker-compose.yml이 있는 폴더로 이동

A screenshot of a Windows PowerShell terminal window. The title bar reads '관리자: Windows PowerShell'. The prompt is 'PS C:\dev\mysql>' and the command 'ls' has been entered. The output shows the directory 'C:\dev\mysql' and a table of files. The table has columns: Mode, LastWriteTime, Length, and Name. There are two entries: a directory named 'database' with a last write time of 2023-11-09 오전 10:25, and a file named 'docker-compose.yml' with a last write time of 2023-10-16 오후 8:07 and a length of 409 bytes.

```
관리자: Windows PowerShell
PS C:\dev\mysql> ls

디렉터리: C:\dev\mysql

Mode                LastWriteTime         Length Name
----                -
d-----          2023-11-09 오전 10:25             database
-a----          2023-10-16 오후 8:07             409 docker-compose.yml

PS C:\dev\mysql>
```

> docker-compose up -d # mysql 생성 및 실행

```
관리자: Windows PowerShell
PS C:\dev\mysql> docker-compose up -d
[+] Running 11/11
 ✓ db 10 layers [#####] 0B/0B Pulled 29.6s
 ✓ 8e0176adc18c Pull complete 2.1s
 ✓ 2d2c52718f65 Pull complete 0.7s
 ✓ d88d03ce139b Pull complete 0.8s
 ✓ 4a7d7f11aa1e Pull complete 1.7s
 ✓ ce5949193e4c Pull complete 1.6s
 ✓ f7f024dfb329 Pull complete 2.3s
 ✓ 5fc3c840facc Pull complete 5.3s
 ✓ 509068e49488 Pull complete 2.9s
 ✓ cbc847bab598 Pull complete 5.8s
 ✓ 942bef62a146 Pull complete 3.7s
[+] Building 0.0s (0/0) docker: default
[+] Running 2/2
 ✓ Network mysql_default Created 0.1s
 ✓ Container mysql-db-1 Started 15.0s
PS C:\dev\mysql> _
```

> docker ps # 생성된 mysql 확인

```
관리자: Windows PowerShell
PS C:\dev\mysql\database> docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
dbd0dc90796b   mysql    "docker-entrypoint.s..." 11 minutes ago Up 11 minutes  0.0.0.0:3306->3306/tcp, 33060/tcp  mysql-db-1
PS C:\dev\mysql\database>
```

> cd database

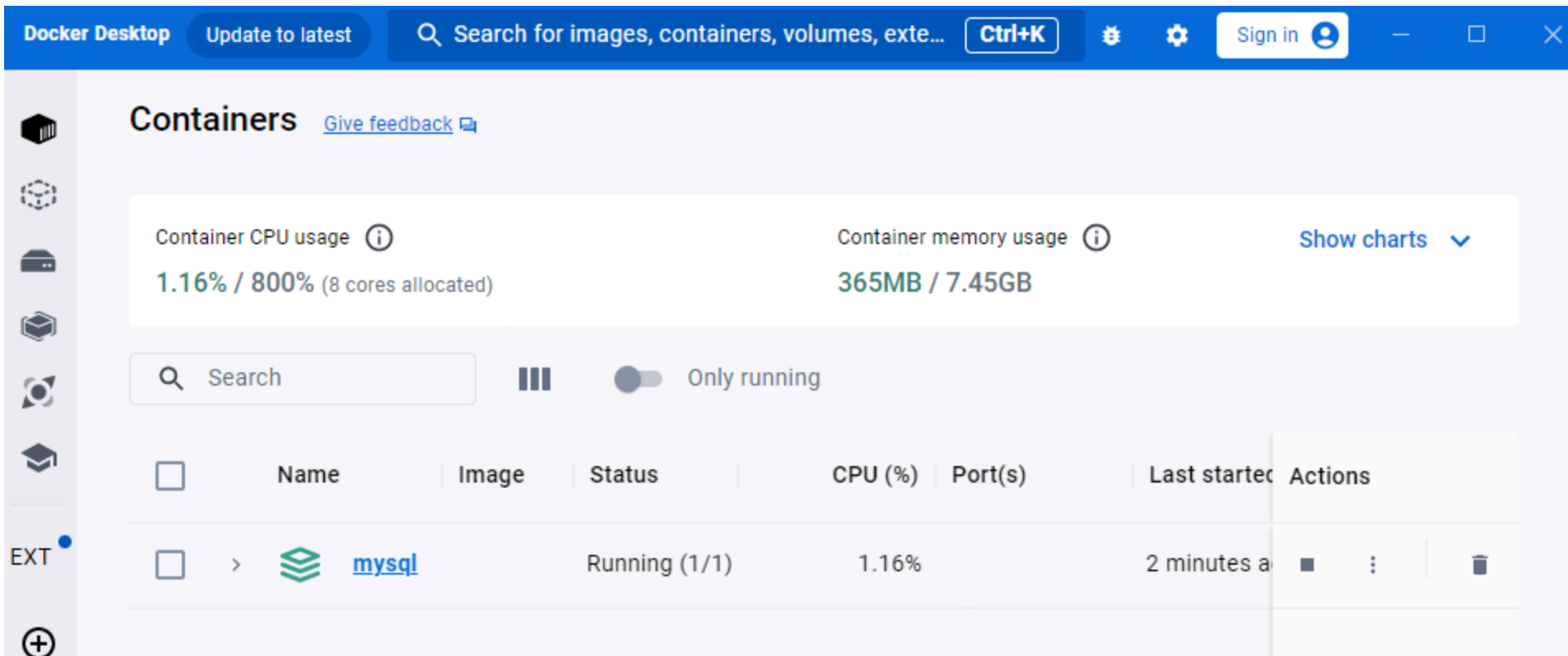
> ls # database 폴더에 mysql이 잘 설치되어 있는지 확인

```
관리자: Windows PowerShell
PS C:\dev\mysql> cd .\database\
PS C:\dev\mysql\database> ls

디렉터리: C:\dev\mysql\database

Mode                LastWriteTime         Length Name
----                -
d-----         2023-11-09 오전 10:31             #innodb_redo
d-----         2023-11-09 오전 10:31             #innodb_temp
d-----         2023-11-09 오전 10:31             examplesdb
d-----         2023-11-09 오전 10:31             mysql
d-----         2023-11-09 오전 10:31             performance_schema
d-----         2023-11-09 오전 10:31             sys
-a-----         2023-11-09 오전 10:31          196608 #ib_16384_0.dblwr
-a-----         2023-11-09 오전 10:31          8585216 #ib_16384_1.dblwr
-a-----         2023-11-09 오전 10:31           56 auto.cnf
```


Docker Desktop에서 mysql 확인



The screenshot shows the Docker Desktop application window. The top bar is blue with the text "Docker Desktop", a button "Update to latest", a search bar "Search for images, containers, volumes, exte...", a "Ctrl+K" button, and icons for Docker, settings, and a user profile with a "Sign in" button. The main area is titled "Containers" with a "Give feedback" link. Below the title, there are two summary cards: "Container CPU usage" showing "1.16% / 800% (8 cores allocated)" and "Container memory usage" showing "365MB / 7.45GB". A "Show charts" link is to the right. Below these cards is a search bar and a toggle switch labeled "Only running". A table lists the containers with columns: Name, Image, Status, CPU (%), Port(s), Last started, and Actions. One container is listed: "mysql" with image "mysql", status "Running (1/1)", CPU "1.16%", and "Last started 2 minutes ago". The Actions column for this container shows a play button, a vertical ellipsis, and a trash icon. On the left sidebar, there are icons for Docker, Kubernetes, Extensions, and a plus sign at the bottom. The "EXT" label is next to the Extensions icon.

Containers [Give feedback](#)

Container CPU usage ⓘ
1.16% / 800% (8 cores allocated)

Container memory usage ⓘ
365MB / 7.45GB

[Show charts](#) ▾

Search

Only running

	Name	Image	Status	CPU (%)	Port(s)	Last started	Actions
☐	>  mysql		Running (1/1)	1.16%		2 minutes ago	  

2. MySQL Community Downloads

[General Availability \(GA\) Releases](#) [Archives](#) [i](#)

MySQL Community Server 8.0.33

Select Operating System:

Microsoft Windows ▼

[Looking for previous GA versions?](#)

Recommended Download:

MySQL Installer for Windows

**All MySQL Products. For All Windows Platforms.
In One Package.**

Starting with MySQL 5.6 the MySQL Installer package replaces the standalone MSI packages.

Windows (x86, 32 & 64-bit), MySQL Installer MSI



[Go to Download Page >](#)

General Availability (GA) Releases

Archives



MySQL Installer 8.0.33

Select Operating System:

Microsoft Windows



[Looking for previous GA versions?](#)

Windows (x86, 32-bit), MSI Installer

8.0.33

2.4M

[Download](#)

(mysql-installer-web-community-8.0.33.0.msi)

MD5: 2a330cf24915964cca87e04dbb34e5d3 | [Signature](#)

Windows (x86, 32-bit), MSI Installer

8.0.33

428.3M

[Download](#)

(mysql-installer-community-8.0.33.0.msi)

MD5: 9b4ce33ab05ae7e0aa30a6c4f1a4d1c2 | [Signature](#)



We suggest that you use the [MD5 checksums](#) and [GnuPG signatures](#) to verify the integrity of the packages you download.

Login Now or Sign Up for a free account.

An Oracle Web Account provides you with the following advantages:

- Fast access to MySQL software downloads
- Download technical White Papers and Presentations
- Post messages in the MySQL Discussion Forums
- Report and track bugs in the MySQL bug system

Login »

using my Oracle Web account

Sign Up »

for an Oracle Web account

MySQL.com is using Oracle SSO for authentication. If you already have an Oracle Web account, click the Login link. Otherwise, you can signup for a free account by clicking the Sign Up link and following the instructions.

No thanks, just start my download.

MySQL Installer

MySQL. Installer
MySQL Server 8.0.33

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Server File Permissions

Apply Configuration

Type and Networking

Server Configuration Type

Choose the correct server configuration type for this MySQL Server installation. This setting will define how much system resources are assigned to the MySQL Server instance.

Config Type: Development Computer

Connectivity

Use the following controls to select how you would like to connect to this server.

☒ TCP/IP

Port: 3306

X Protocol Port: 33060

☒ Open Windows Firewall ports for network access

☐ Named Pipe

Pipe Name: MYSQL

☐ Shared Memory

Memory Name: MYSQL

Advanced Configuration

Select the check box below to get additional configuration pages where you can set advanced and logging options for this server instance.

☐ Show Advanced and Logging Options

Next >

Cancel

MySQL Installer

MySQL Server 8.0.33

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Server File Permissions

Apply Configuration

Authentication Method

☐ Use Strong Password Encryption for Authentication (RECOMMENDED)

MySQL 8 supports a new authentication based on improved stronger SHA256-based password methods. It is recommended that all new MySQL Server installations use this method going forward.



Attention: This new authentication plugin on the server side requires new versions of connectors and clients which add support for this new 8.0 default authentication (caching_sha2_password authentication).

Currently MySQL 8.0 Connectors and community drivers which use libmysqlclient 8.0 support this new method. If clients and applications cannot be updated to support this new authentication method, the MySQL 8.0 Server can be configured to use the legacy MySQL Authentication Method below.

☒ Use Legacy Authentication Method (Retain MySQL 5.x Compatibility)

Using the old MySQL 5.x legacy authentication method should only be considered in the following cases:

- If applications cannot be updated to use MySQL 8 enabled Connectors and drivers.
- For cases where re-compilation of an existing application is not feasible.
- An updated, language specific connector or driver is not yet available.

Security Guidance: When possible, we highly recommend taking needed steps towards upgrading your applications, libraries, and database servers to the new stronger authentication. This new method will significantly improve your security.

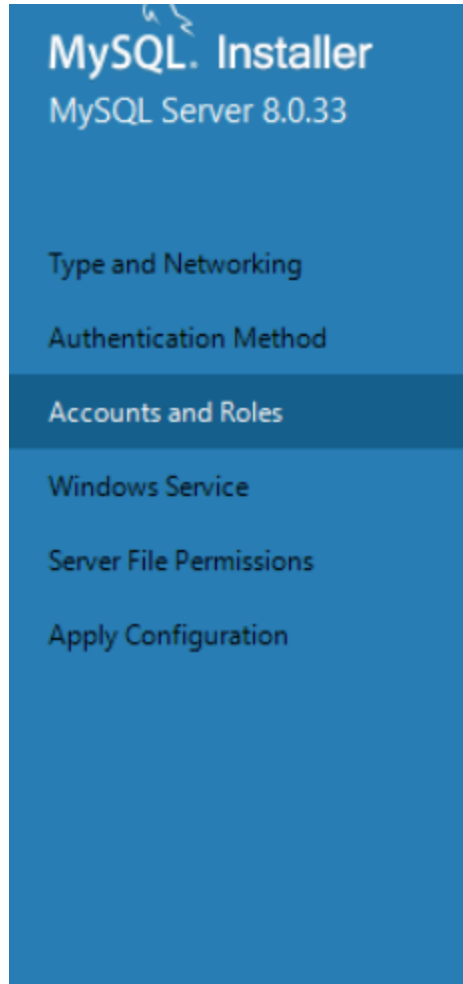
< Back

Next >

Cancel

관리자 계정의 비번 생성

- 관리자 계정은 Root입니다.



Accounts and Roles

Root Account Password

Enter the password for the root account. Please remember to store this password in a secure place.

MySQL Root Password:

Repeat Password:

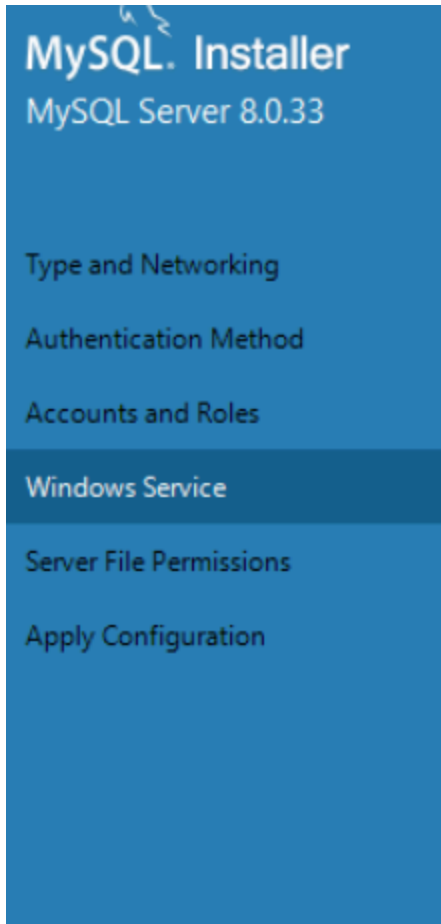
Password strength: **Weak**

MySQL User Accounts

Create MySQL user accounts for your users and applications. Assign a role to the user that consists of a set of privileges.

MySQL User Name	Host	User Role	
			Add User Edit User Delete

윈도우즈(OS)에 서비스 등록



Windows Service

☒ Configure MySQL Server as a Windows Service

Windows Service Details

Please specify a Windows Service name to be used for this MySQL Server instance.
A unique name is required for each instance.

Windows Service Name:

☒ Start the MySQL Server at System Startup

Run Windows Service as ...

The MySQL Server needs to run under a given user account. Based on the security requirements of your system you need to pick one of the options below.

☒ Standard System Account

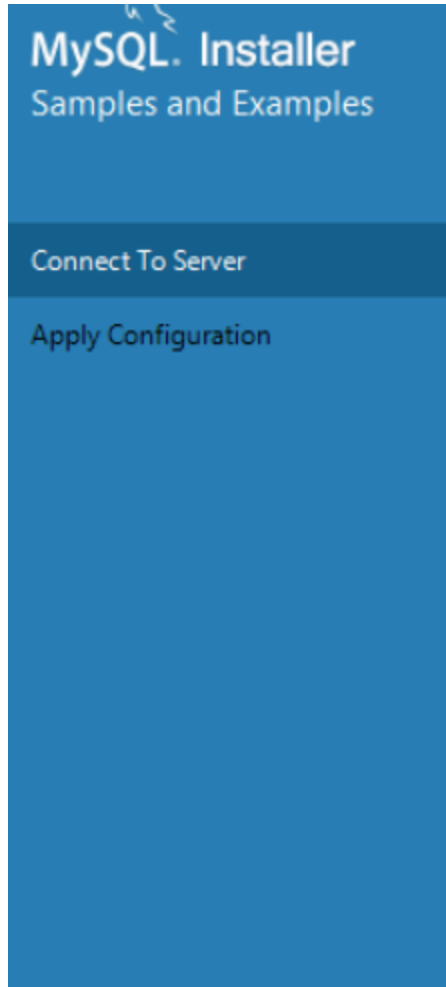
Recommended for most scenarios.

☐ Custom User

An existing user account can be selected for advanced scenarios.

Connect To Server

- 관리자 계정(root)로 접속



Connect To Server

Select the MySQL server instances from the list to receive sample schemas and data.

	Server	Port	Arch...	Type	Status
☒	MySQL Server 8.0.33	3306	X64	Stand-alone Server	Connection succeeded.

Provide the credentials that should be used (requires root privileges).
Click "Check" to ensure they work.

User name: Credentials provided in Server configuration

Password:



MySQL. Installer

Adding Community

Choosing a Setup Type

Check Requirements

Installation

Product Configuration

Installation Complete

Installation Complete

The installation procedure has been completed.

Copy Log to Clipboard

☐ Start MySQL Workbench after setup

☐ Start MySQL Shell after setup

The MySQL Shell is an advanced MySQL client application that can be used to work with single MySQL Server instances. Further, it can be used to create and manage InnoDB Cluster, an integrated solution for high availability and scalability of MySQL databases, without requiring advanced MySQL expertise.



Refer to the following links for documentation, tutorials and examples on MySQL Shell:

[MySQL Shell Documentation](#)

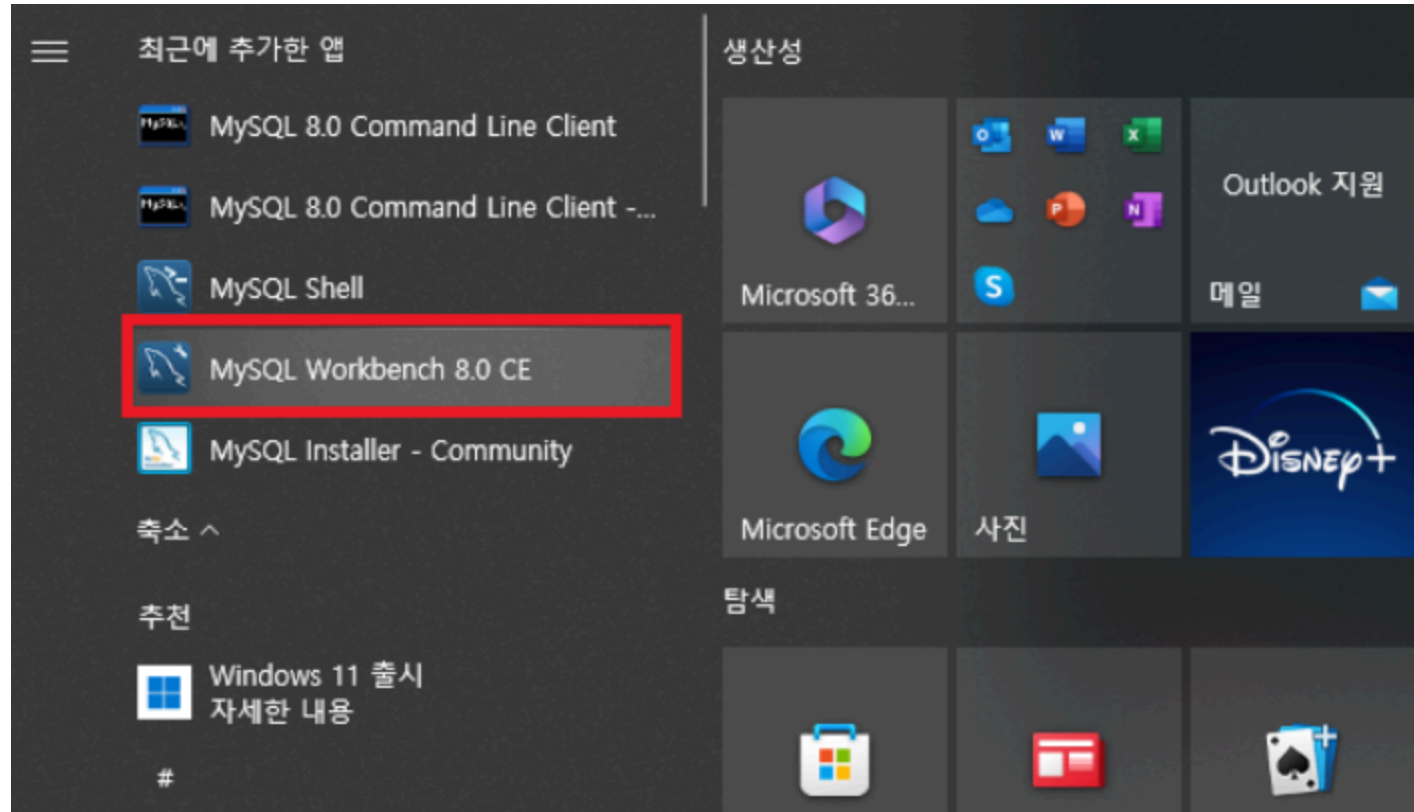
[Setting up a Real World Cluster Blog](#)

[The All New MySQL InnoDB ReplicaSet Blog](#)

[Changing Cluster Options Live Blog](#)

Finish

설치 결과 확인



MySQL Server 구동방법

```
> mysql.server start # 서버 시작  
> mysql.server stop # 서버 멈춤  
> mysql.server restart # 서버 재시작  
> mysql.server status # 서버 상태확인
```